multiples of the Planck length, if one circle has radius *R*, the mathematical dictionary shows that the other circle has radius 1/*R*). You might think that you could easily and immediately distinguish between a big and a small dimension, but in string theory this is not always the case. All observations derive from the interactions of strings, and these two theories, the Type IIA with a big circular dimension and the Type IIB with a small circular dimension, are merely different translations of —different ways of expressing—the same physics. Every observation you describe within one string theory has an alternative and equally viable description within the other string theory, even though the language of each theory and the interpretation it gives may differ. (This is possible because there are two qualitatively different configurations for strings moving on a circular dimension: those in which the string is wrapped around the circle like a rubber band around a tin can, and those in which the string resides on a portion of the circle but does not wrap around it. The former have energies that are *proportional* to the radius of the circle [the larger the radius, the longer the wrapped strings are stretched, so the more energy they embody], while the latter have energies that are *inversely proportional* to the radius [the smaller the radius, the more hemmed in the strings are, so the more energetically they move because of quantum uncertainty]. Notice that if we were to replace the original circle by one of *inverted* radius, while also exchanging "wrapped" and "not wrapped" strings, physical energies—and, it turns out, physics more generally—would remain unaffected. This is exactly what the dictionary translating from the Type IIA theory to the Type IIB theory requires, and why two seemingly different geometries—a big and a small circular dimension—can be equivalent.)

A similar idea also holds when circular dimensions are replaced with the more complicated Calabi-Yau shapes introduced in Chapter 12. A given string theory with extra dimensions curled up into a particular Calabi-Yau shape gets translated by the dictionary into a different string theory with extra dimensions curled up into a different Calabi-Yau shape (one that is called the *mirror* or *dual* of the original). In these cases, not only can the sizes of the Calabi-Yaus differ, but so can their shapes, including the number and variety of their holes. But the translation dictionary ensures that they differ in just the right way, so that even though the extra dimensions have different sizes and shapes, the physics